MBA: 8109 RESEARCH METHODS



Session 10 — Analysing Quantitative Data

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LEARNING OBJECTIVES

By the end of this session, the candidate should be able to

1. **Prepare, code and check** quantitative data so it is ready for analysis.
2. **Select appropriate statistics** — descriptive and inferential — for a given research question.
3. **Interpret statistical output** in relation to the research objectives, not in isolation.
4. **Relate techniques to practice** — the designs seen in the dissertations reviewed in the Week 3 syndicate exercise (Section 8).

Where this lands in your work

- Concept Paper, Chapter 3: data analysis plan
- Final dissertation, Chapter 4: presentation of findings (SU Guide)

Analysis starts with your objectives, not your software

1. Prepare & code
Data matrix, codebook, missing-value codes

2. Check & clean
Errors, duplicates, outliers, reliability

3. Explore & describe
Tables, charts, central tendency, spread

4. Test & model
Associations, differences, relationships

5. Interpret
Answer each objective; link to theory

Plan it before you collect it

Decide scale of measurement, input format, coding, weighting and error-checking methods before data arrive (Saunders et al., Ch. 12).

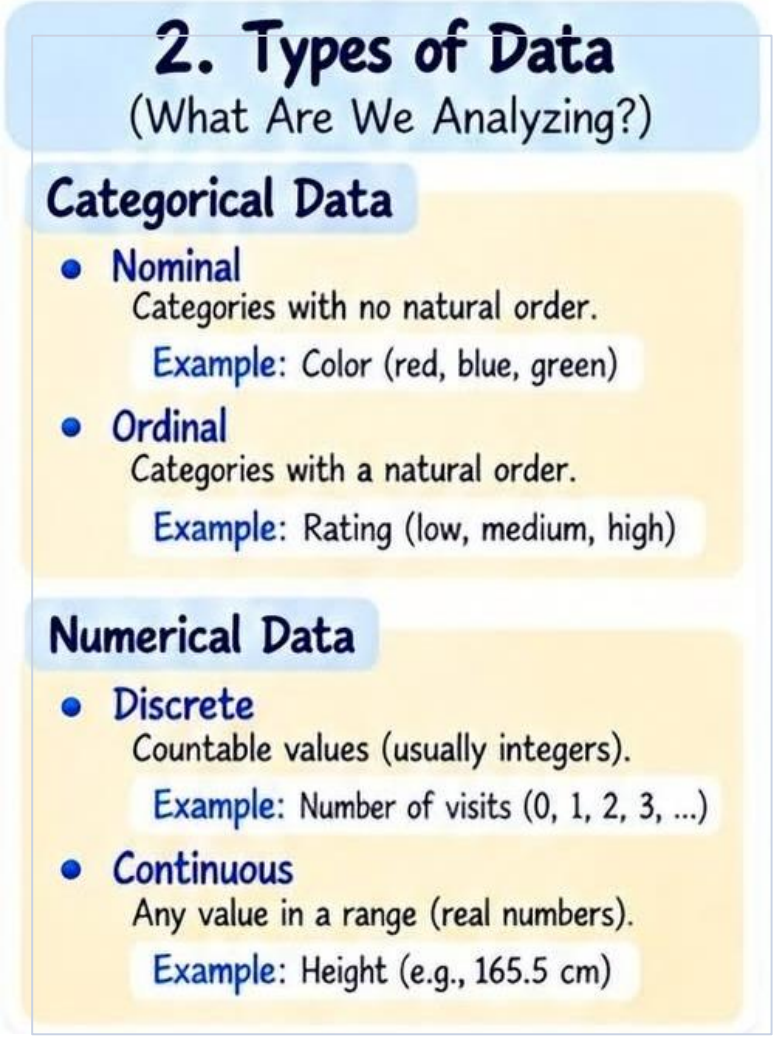
Rubbish in, rubbish out

Software makes it easy to produce elegantly presented nonsense. Careful preparation is what makes the output trustworthy.

The golden thread

Objective → variable & construct (Ch. 2 operationalisation table) → scale → statistic → interpretation. Break one link and the analysis fails.

The scale of measurement constrains every choice that follows



Visual: course infographic (Data Analysis)

Scale (Saunders)	Business example	Describe with	Typical tests	Present with
Nominal (descriptive)	Sector, gender, region	Frequencies, %, mode	Chi-square, Cramér's V, phi	Bar or pie chart
Ordinal (ranked)	Single Likert item, job grade	Median, IQR	Spearman, Mann-Whitney, Kruskal-Wallis	Bar chart
Interval	Composite Likert scale score*	Mean, SD	t-test, ANOVA, Pearson, regression	Histogram, box plot
Ratio	Revenue, tenure, age, sales	Mean, SD, CV	As interval; index numbers for trends	Histogram, line, scatter

***Defend your assumption.** A single Likert item is ordinal. Summed or averaged multi-item scales are conventionally treated as interval — state this in Chapter 3 and check reliability first. Interval and ratio data can always be grouped downwards into categories; categorical data can never be upgraded.

Build a clean data matrix before you run a single test

1 Data matrix
Rows are cases, columns are variables. The first column is a unique case ID linked to the original questionnaire.

2 Pre-code the instrument
Put numeric codes on closed questions. For open questions, build a codebook from the first 50–100 cases.

3 Code every cell
Including missing data. deVaus's reasons: not required (filter skip), refused, don't know, missed in error.

4 Label and transform
Variable and value labels mirror instrument wording. Reverse-code negative items; compute composite scores; weight strata if needed.

Extract from a coded data matrix

ID	Sector	Tenure	TL1	TL2	TL3r	TL_mean
001	1	4	5	4	4	4.33
002	3	12	3	3	2	2.67
003	2	99	4	8	4	4.00
004	1	7	2	3	9	2.50

Codebook notes

- Sector: 1 = Manufacturing, 2 = Services, 3 = Public
- TL1–TL3: 1 = Strongly disagree … 5 = Strongly agree; TL3 reverse-coded (TL3r)
- Missing: 99 = not stated, 8 = don't know, 9 = refused — declared as missing in software
- TL_mean averages valid items only

Every dataset has errors — find them before your examiner does

Saunders's three error checks

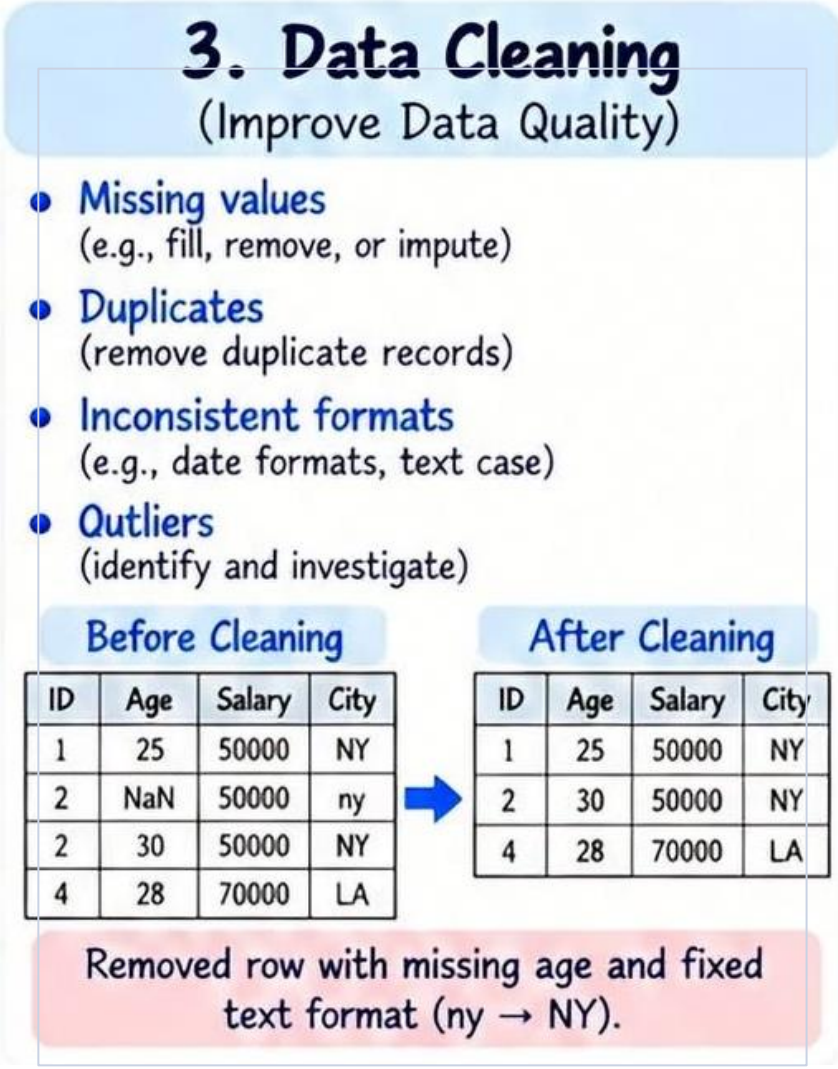
Illegitimate codes
A 7 on a 1–5 scale; letter O typed for zero.

Illogical relationships
A 'senior manager' with one month's tenure.

Filter rules broken
Answers recorded where the filter said skip.

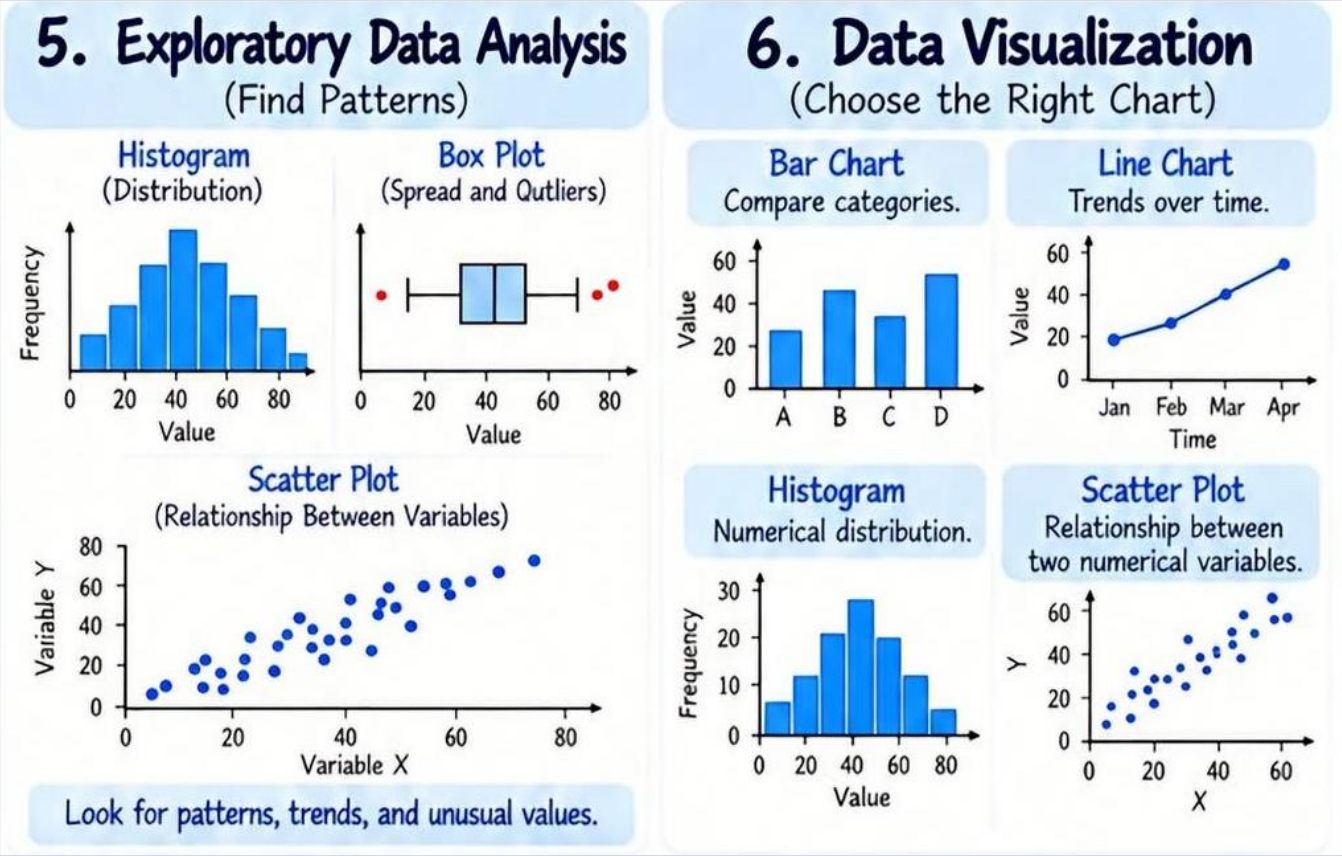
Then screen and prepare

- Duplicates and inconsistent formats (dates, text case) — remove or standardise.
- Missing data — report the pattern; choose listwise, pairwise or imputation and justify it.
- Outliers — investigate with box plots and z-scores beyond ±3; correct errors, keep genuine cases.
- Reliability — Cronbach's alpha of 0.70 or above before you compute composite scores.
- Record response rate and every cleaning decision for Chapter 4.



Visual: course infographic (Data Analysis)

Choose the table or chart by what you want the reader to see



Visual: course infographic (Data Analysis) — exploratory analysis and visualisation

To show…	Use
Specific values	Table / frequency distribution
Highest and lowest	Bar chart (categorical); histogram (numerical)
Trend over time	Line graph
Proportions	Pie chart; percentage component bar
Distribution and outliers	Histogram; box plot
Two variables together	Cross-tabulation; multiple bar chart
Relationship between two numerical variables	Scatter graph

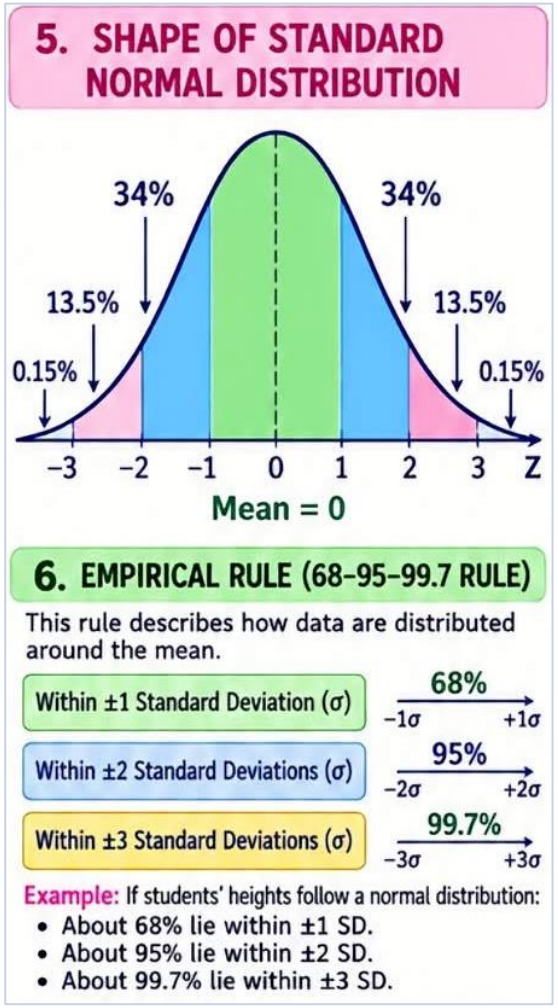
Every table and figure needs: a clear title, units, data source, sample size (n), labelled axes or headings, and a legend where needed. Avoid 3-D charts — they distort.

Centre, spread and shape — and why the normal curve matters

Central tendency
Mean (numerical, symmetric) · Median (skewed or ranked) · Mode (categorical)

Dispersion
Range · Inter-quartile range (with median) · Standard deviation (with mean) · Coefficient of variation to compare groups

Shape
Skewness and kurtosis; check normality with a histogram, Q-Q plot, Kolmogorov-Smirnov or Shapiro-Wilk



Visual: course infographic (Standard Normal Distribution)

3. Z-SCORE FORMULA

A value from any normal distribution can be converted into a Z-score.

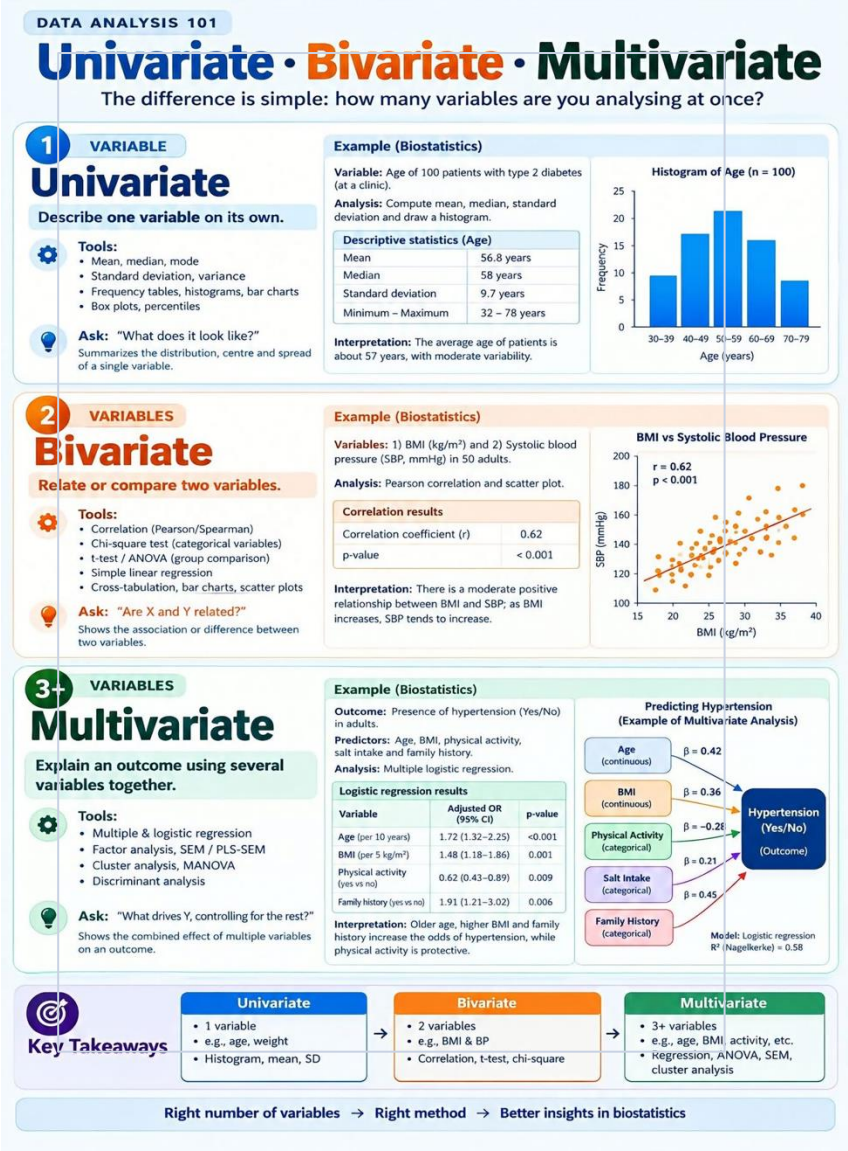
$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = Observed value
- μ = Population mean
- σ = Population standard deviation
- z = Standard score

Worked example: mean = 70, SD = 10, score = 85 → z = (85 − 70) / 10 = 1.5 — one and a half standard deviations above the mean.

How many variables does your objective involve?



Visual: course infographic (Data Analysis 101)

Univariate — one variable *"What does it look like?"*

Level of employee engagement in Kenyan banks: mean, SD, frequency table, histogram.

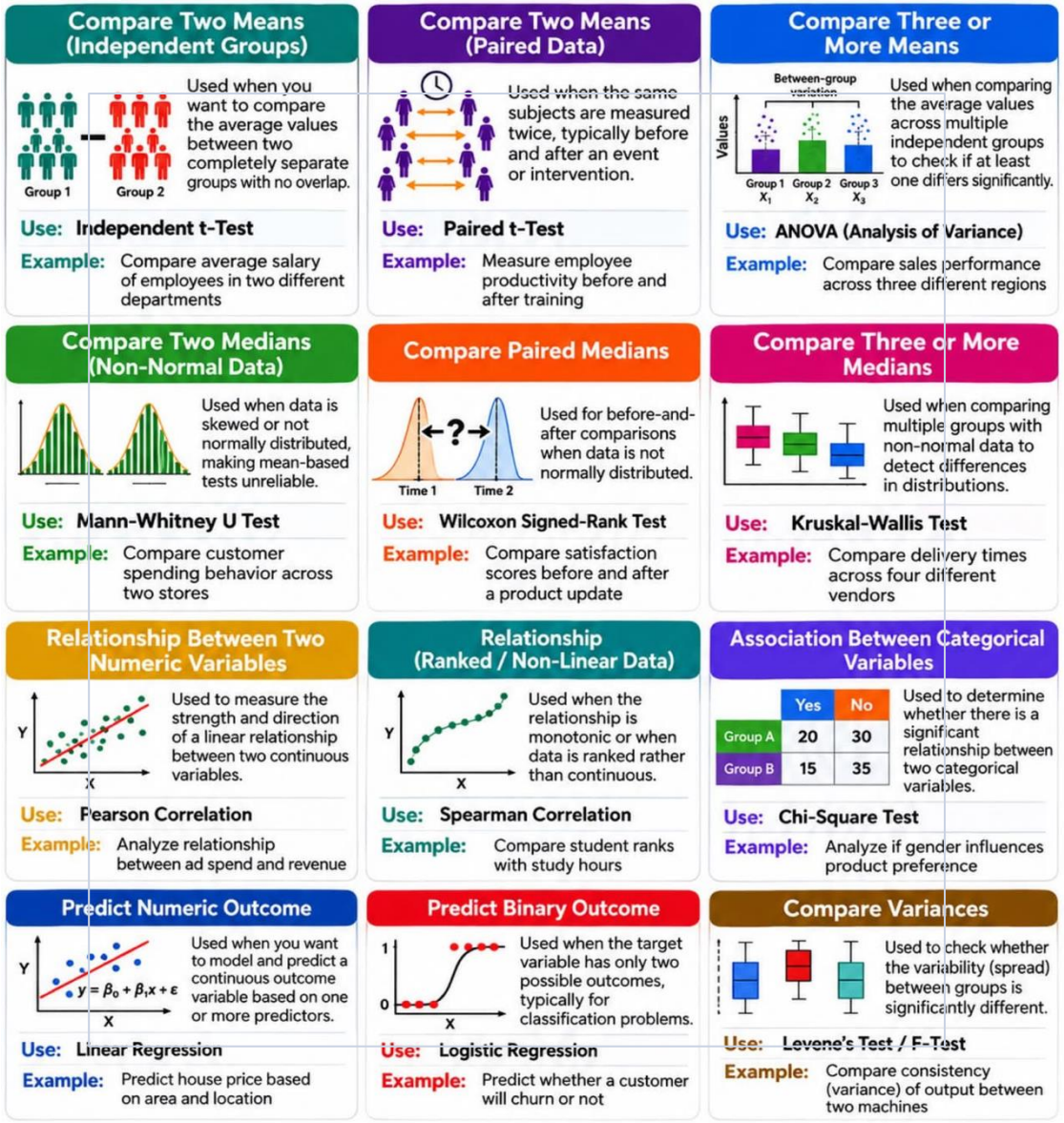
Bivariate — two variables *"Are X and Y related or different?"*

Transformational leadership and engagement: Pearson r; engagement by sector: ANOVA; gender by grade: chi-square.

Multivariate — three or more *"What drives Y, controlling for the rest?"*

Firm performance on four strategy dimensions, controlling for firm size: multiple regression. Moderators and mediators: hierarchical regression, PROCESS or SEM.

Which statistical test should I use?



Four questions that choose the test

1. What is the goal — association, difference, relationship or prediction?
2. What scale are the dependent and independent variables?
3. How many groups, and are they independent or paired?
4. Are parametric assumptions met?

Parametric assumptions

- Independent cases
- Normally distributed populations
- Equal variances (Levene's test)
- Numerical data

Not met? Use the non-parametric equivalent.

Significance testing

H_0: no association or difference. $p \leq 0.05 \rightarrow$ reject H_0.

Type I: rejecting a true H_0 — convicting the innocent.

Type II: keeping a false H_0 — acquitting the guilty.

Large samples make trivial effects significant.

Visual: course infographic (Which Statistical Test Should I Use?)

Read output back against the objective it was meant to answer

Objective	Hypothesis	Test	Report the output	Interpretation for the objective
Determine the effect of transformational leadership on employee engagement	H_{o1}: no significant effect	Simple linear regression	$\beta = 0.46$, $t = 6.10$, $p < .001$, $R^2 = .21$	Reject H_{o1}. Leadership explains about 21% of variation in engagement — consistent with the theory reviewed in Chapter 2.
Examine whether engagement differs by sector	H_{o2}: no difference between sectors	One-way ANOVA + Tukey post hoc	$F(2, 197) = 4.37$, $p = .014$, $\eta^2 = .04$	Reject H_{o2}. Services score higher than public sector, but the effect is small.
Establish whether grade is associated with gender	H_{o3}: no association	Chi-square + Cramér's V	$\chi^2(4) = 33.59$, $p < .001$; $V = .295$	Reject H_{o3}. A significant but weak association: men are more likely to hold higher grades.

Report fully
Statistic, df, p, effect size and, where possible, confidence intervals. Software's p = .000 means p < .001.

Significant ≠ important
Judge practical meaning from effect size and context, not the p-value alone.

Association ≠ causation
Cross-sectional surveys show relationships; avoid causal verbs unless the design supports them.

Match the software to your data and your analysis

Tool	Type	Best for	Typical use in an MBA dissertation
Excel	Spreadsheet, Analysis ToolPak	Data entry, cleaning, descriptives, pivot tables, charts	Building the data matrix; frequencies and cross-tabs
SPSS	Menu-driven statistics package	t-tests, ANOVA, chi-square, correlation, regression, Cronbach's alpha	Questionnaire surveys with Likert scales
Stata / EViews	Command-based econometrics	Panel, time-series and secondary data; logged do-files	Firm, bank or country data over several years
R (RStudio)	Free, open-source code	Any technique, from descriptives to advanced models; graphics	Analyses beyond menu packages; reproducible scripts
Python	Free, general-purpose code (pandas, statsmodels)	Large datasets, automation, machine learning	Web, transactional or social media data
SmartPLS / AMOS	Structural equation modelling	Latent constructs, mediation and moderation	Testing a conceptual framework of multi-item constructs

Choose the test, then the tool

Pick the statistic first (slides 9–10). Any of these packages will run a t-test or regression.

Free or licensed

R, Python and JASP are free. SPSS, Stata, AMOS and SmartPLS need a licence; check what your institution provides.

Keep it reproducible

Save syntax, do-files or scripts so an examiner can trace every result in Chapter 4.

Audit the Week 3 dissertations, then run the statistics yourself

Syndicate audit (Week 3, Section 8)

For each dissertation your group reviewed:

- Map each objective to the test actually used.
- Were the data screened and assumptions reported?
- Does the chart suit the data type and question?
- Does the interpretation return to the objective and theory?
- Red flags: Pearson on single Likert items; chi-square with many expected counts below 5; regression without diagnostics; causal claims from cross-sectional data.

statistical practical

1. **Load and inspect:** Open the sample dataset and codebook; declare missing values.
2. **Screen:** Find illegitimate codes and outliers; reverse-code; compute alpha and composites.
3. **Describe:** Frequencies, mean and SD, histogram and box plot; check normality.
4. **Test:** Cross-tab with chi-square, t-test, ANOVA, Pearson or Spearman, simple and multiple regression.
5. **Write up:** One reporting paragraph, one table and one chart per objective.

Concept paper: what to submit and how

1 Follow the Strathmore guidelines

Structure, formatting and referencing must follow the Strathmore University guidelines on research papers.

Declaration at the foot of the title page: original work, name printed, AI use declared.

2 Submit two versions

- **MS Word version** uploaded through the system. 1300hrs for exam day EAT. Note on AI Checker and Similarity Policy
- **Bound hard copy** handed in on the exam day, when you sign the examination nominal roll

No examination can be graded if you have not signed the examination nominal roll. Any use of AI must be declared

3 About 20 pages of content, in this order

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Ends with declaration

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Sample data collection tools